A

Project Report

On

KisanHaat: A Web-Based Platform for Agricultural Product Trading and Machinery Rental

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DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that Project Report entitled "KisanHaat: A Web-Based Platform for Agricultural Product Trading and Machinery Rental" which is submitted by "Amresh Prajapati (2200291530010), Abhinav Kumar (2200291530002), Aditya Yadav (2200291530007)" in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ABSTRACT

The agricultural sector in India faces several challenges in marketing farm produce and accessing agricultural machinery at affordable costs. Farmers often struggle to find fair buyers for their crops, while also facing difficulties in arranging machinery for short-term agricultural activities. The absence of a centralized digital platform leads to dependency on middlemen, price exploitation, and inefficient resource utilization.

To address these challenges, we developed KisanHaat, a full-stack web application that serves as a digital marketplace for buying and selling agricultural products and renting agricultural machinery. The platform connects farmers, buyers, and machinery owners in a transparent and secure environment. Farmers can list their agricultural products for sale, buyers can search and purchase products based on their needs, and machinery owners can post equipment for rental purposes.

The frontend of KisanHaat is developed using React.js to provide an interactive and responsive user interface. The backend is implemented using Node.js and Express.js to manage business logic and API services, while MongoDB is used as the database to store user details, product listings, rental data, and transaction records. The system uses phone number-based authentication with OTP verification to ensure secure access. Additionally, Google Maps API integration enables users to filter products and machinery based on location, improving accessibility and convenience.

The platform includes a structured rental request and approval mechanism, where users can add machinery to their cart and send rental requests, which must be approved by the owner before confirmation. This ensures transparency and mutual trust between users.

The implementation and testing of KisanHaat demonstrate that the system simplifies agricultural trade, reduces dependency on intermediaries, and promotes digital transformation in rural markets. The proposed solution provides a scalable, secure, and user-friendly platform that supports both agricultural commerce and machinery rental services.

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LIST OF ABBREVIATIONS

API Application Programming Interface
UI User Interface
SDGs Sustainable Development Goals

SDG MAPPING

Kisanhaat: An Online Agricultural Trading of products and farming equipment is in line with the global development framework that is developed by the United Nations under Sustainable Development Goals (SDGs). In India, agriculture is a significant factor in the stability of the economy, food security and rural employment. Nevertheless, farmers are usually undergoing some issues like their inability to access the market directly, reliance on intermediaries, inadequate access to digital infrastructure, and disproportional access to agricultural resources.

Kisanhaat is a contributor to sustainable agricultural development, offering an efficient and transparent platform technology-driven that links farmers, buyers as well as machine owners. The system promotes digital inclusiveness, fair trade, and optimization of resources among the rural communities. The suggested platform is directly related to SDG 2 (Zero Hunger), SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure) and SDG 10 (Reduced Inequalities).

CHAPTER 1

INTRODUCTION

1.1 Introduction

One of the most significant divisions of the Indian economy is agriculture that employs a significant part of the population, produces food, and develops rural areas. A significant proportion of the population is either directly or indirectly reliant upon farming as a source of livelihood. Even in the face of technology in other sectors, a number of agricultural markets are still in the traditional and disorganized mode of operation.

Farmers have been experiencing various problems including the absence of direct contact with buyers, relying on middlemen, exploitation of prices, and late payments. Equally, farm machinery needed to carry out farming activities such as ploughing, harvesting, and irrigation is costly and cannot be afforded by small and marginal farmers. This makes most farmers rent machines either informally or they have to rely on local contacts which is not very reliable and efficient.

With the rapid growth of digital technology and smartphone penetration in rural areas, there is a strong opportunity to digitize agricultural trade and machinery rental systems. A centralized digital platform can bring transparency, reduce exploitation, and improve accessibility for farmers and buyers.

These challenges and opportunities form the foundation for the development of **Kisanhaat,** a web-based digital marketplace for agricultural product trading and machinery rental.

1.2 Project Description

KisanHaat: A Web-Based Platform for Agricultural Product Trading and Machinery Rental is designed to make life easier for farmers by bringing agricultural trading and machinery sharing onto a single digital platform. In many rural areas, farmers struggle and are able to get fair prices for their crops due to involvement of middlemen. At the same time, purchasing expensive machinery is not affordable for every farmer, and renting equipment is often difficult to arrange.

KisanHaat aims to solve both of these challenges through a simple and practical web-based solution.

The integration of digital platforms in agriculture can significantly improve efficiency, transparency, and accessibility. A web-based system can allow farmers to directly list their agricultural products for sale and connect with buyers without intermediaries.

Digital solutions offer several advantages:

- Real-time product listing and updates
- Wider market reach beyond local mandis
- Transparent communication between buyers and sellers
- Easy comparison of products and rental prices
- Location-based filtering using mapping technologies

By leveraging use of modern web technologies such as React.js used for frontend development and Node.js with Express.js used for backend services, it becomes possible to build scalable and farmer-friendly systems designed for agricultural use.

1.3 Problems with Traditional Agricultural System

The traditional agricultural marketing system is largely dependent on physical mandis (local markets) and intermediaries. Farmers transport their produce to local markets where prices are often determined by middlemen rather than fair market competition. This results in reduced profit margins for farmers.

Some major limitations of the traditional system include:

- Lack of price transparency
- Dependence on intermediaries
- Limited market reach
- Delayed payments
- High transportation costs

Similarly, renting agricultural machinery through traditional methods also presents challenges. Farmers usually rely on local contacts or offline negotiations. There is no structured system for availability checking, price comparison, or booking confirmation. Additionally, there is no formal approval mechanism to ensure transparency between machinery owners and renters.

These limitations highlight the need for a digital and structured platform to streamline agricultural trade and machinery rental processes.

1.4 Limitations of Existing Digital Platforms

Although some agricultural marketplaces exist, many platforms do not fully integrate both product trading and machinery rental in a single system. Some platforms focus only on crop selling, while others focus solely on equipment rental.

Common limitations of existing systems include:

- Complex user interfaces unsuitable for rural users
- Lack of proper authentication mechanisms
- No structured rental approval process
- Limited location-based filtering
- Inadequate trust-building mechanisms

Moreover, many platforms lack a proper cart-based system for machinery rental requests and do not ensure mutual consent between renters and machinery owners.

These gaps indicate the need for a more structured and farmer-friendly digital solution.

1.5 Motivation of the Project

The primary motivation behind developing **Kisanhaat** is to empower farmers by providing a transparent and accessible digital marketplace. Many farmers, especially in rural areas, face

financial losses due to unfair pricing and lack of direct buyer access. Additionally, the high cost of purchasing agricultural machinery creates a financial burden.

By creating a platform where:

- Farmers can sell their agricultural products directly
- Buyers can search and purchase products easily
- Machinery owners can list equipment for rent
- Users can request rentals through a structured approval system

the project aims to reduce middleman dependency and promote fair trade for better pricing of the crops.

The increasing availability of smartphones and internet connectivity in rural India further motivated the development of a web-based agricultural solution that is simple, secure, and scalable.

1.6 Problem Statement

Farmers often face challenges in selling agricultural products at fair prices due to dependence on intermediaries and limited market access. Additionally, small and marginal farmers struggle to access agricultural machinery for short-term use because purchasing such equipment is financially unfeasible.

Existing traditional and digital systems do not provide a fully integrated, secure, and user-friendly platform that combines agricultural product trading with machinery rental services. Therefore, there is a need to develop a scalable digital platform that ensures transparency, secure user authentication, location-based search, and structured rental approval mechanisms.

The proposed system, Kisanhaat, focuses on solving these issues by providing a comprehensive web-based solution for agricultural commerce and machinery rental.

1.7 Objectives of the Project

The main objectives of the Kisanhaat project are as follows:

1. To develop a digital marketplace for buying and selling agricultural products.
2. To create a structured platform for renting agricultural machinery.
3. To implement secure phone number-based authentication with OTP verification.
4. To provide location-based filtering using Google Maps integration.
5. To implement a cart-based rental request system with owner approval.
6. To ensure secure data storage using MongoDB database.
7. To create a responsive and easy-to-use interface using React.js.

1.8 Scope of the Project

Kisanhaat covers the design, development, and launch of a full-stack web application to facilitate agricultural product trading and machinery rental services.

The key areas of scope are:

1. Development of User Authentication System
2. Implementation of secure login and registration using phone number and OTP verification.
3. Product Listing and Management
4. Allowing farmers to add, update, and manage agricultural product listings.
5. Machinery Rental Module
6. Providing machinery owners with the ability to list equipment for rent and manage rental requests.
7. Cart-Based Rental Request System
8. Users can add machinery to a cart and send rental requests, which must be approved by the owner before confirmation.
9. Location-Based Filtering

10. Integration of Google Maps API to enable users to search products and machinery based on geographical location.
11. Database Management
12. Using MongoDB to securely store user details, product listings, machinery data, and transaction records.
13. Performance and Usability Evaluation
14. Testing system functionality, response time, database efficiency, and user experience to ensure reliability and scalability.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

The rapid rise of digital platforms has greatly changed traditional business models across various sectors, including agriculture. However, agricultural trade in many developing regions still relies on conventional mandi-based systems and informal rental mechanisms. Farmers often face issues such as price exploitation, lack of transparency, limited market access, and inefficient resource utilization.

With advancements in web technologies, cloud databases, and mobile accessibility, several digital agricultural marketplaces and rental systems have been proposed. These platforms aim to improve connectivity between farmers, buyers, and equipment owners. However, many existing systems lack integration, scalability, secure authentication mechanisms, and structured rental approval workflows.

This chapter reviews existing research and digital platforms related to agricultural e-commerce, online rental systems, and secure web application architectures. The objective of this review is to analyze current methodologies, identify their strengths and limitations, and highlight the research gaps that motivated the development of Kisanhaat.

2.2 Review of Existing Literature

2.2.1 e-NAM (National Agriculture Market)

The e-NAM (National Agriculture Market) is an initiative by the Government of India that aims to connect agricultural mandis through an online trading platform. It enables farmers to check real-time prices and sell their produce digitally.

The system improves transparency and price discovery. However, it still depends heavily on mandi-level integration and does not provide a complete direct buyer-to-farmer marketplace experience. Additionally, machinery rental services are not integrated into the same platform.

This limitation highlights the need for a more unified digital solution combining agricultural product trading and machinery rental, as proposed in Kisanhaat.

2.2.2 DeHaat Platform

DeHaat is a privately owned agritech model, which provides a full agro solution, such as supplying inputs, providing technical advice, and linking farmers to the market.

While DeHaat focuses on supply chain efficiency and farmer support, its services are more enterprise-driven rather than a fully open marketplace where individual farmers and buyers can directly interact. Furthermore, rental management with structured approval systems is not the primary focus.

This shows the need for a simplified, farmer-centric digital marketplace like Kisanhaat, designed specifically for ease of use and direct transactions.

2.2.3 Online Rental System Models

(a) Several studies have explored generic online rental platforms used in industries such as vehicle rental, property rental, and equipment leasing. These systems typically include:

(b) Item listing modules

(c) Booking mechanisms

(d) Availability tracking

(e) User authentication

(f) Payment integration

(g) However, most generic rental systems do not address agricultural-specific challenges such as seasonal demand, rural accessibility, and location-based filtering in low-resource settings.

(h) Kisanhaat adapts the rental model specifically for agricultural machinery, integrating cart-based requests and owner approval mechanisms to ensure transparency and trust between farmers and equipment owners.

2.2.4 Secure Authentication in Web Applications

Modern web applications increasingly rely on secure authentication mechanisms such as OTP verification, JWT tokens, and encrypted databases. Studies in web security emphasize the need for multi-factor authentication to enhance security and avoid unauthorized access.

OTP-based authentication using a registered phone number has proven effective in regions where email access is limited but mobile connectivity is widespread. This is particularly relevant in rural India, where mobile penetration is significantly higher than digital literacy levels.

Kisanhaat incorporates phone number-based OTP verification to ensure secure and user-friendly authentication suitable for rural users.

2.2.5 Location-Based Services in Web Platforms

Location-based filtering and mapping technologies have become essential components of modern digital marketplaces. Integration of mapping services enables users to:

- Search nearby listings
- Filter based on geographical proximity
- Improve logistics planning

Many agricultural studies suggest that proximity-based systems reduce transportation costs and improve transaction efficiency.

Kisanhaat integrates Google Maps API to enable location-based search of products and machinery, making it easier for farmers and buyers to connect locally.

2.2.6 E-Commerce Database Design using MongoDB

The current work in web application development emphasizes the significance of scalable and flexible databases. NoSQL databases such as MongoDB have been prolific in marketplace applications because of:

- Schema flexibility
- Easy scalability
- Efficient handling of large datasets
- Fast read/write operations

Studies show that MongoDB is suitable for dynamic applications where product listings and rental records frequently change. This supports the architectural choice made in Kisanhaat for managing users, products, and rental transactions efficiently.

2.3 Research Gaps

Based on the review of existing systems and literature, the following gaps are identified:

1. Most agricultural platforms focus either on product trading or advisory services, but not on integrated machinery rental systems.
2. Many systems lack a structured rental approval mechanism ensuring mutual consent.
3. Limited emphasis on simple user interfaces tailored for rural users.

4. Inadequate integration of cart-based rental workflows in agricultural contexts.

5. Lack of unified platforms combining product marketplace and machinery rental services.

These loopholes show that there is a necessity of a secure, integrated, and user friendly agricultural market place system like Kisanhaat.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 Overview of Proposed System

KisanHaat: An online Agricultural Product Trading and Renting Machinery is aimed at simplifying the life of farmers by allowing them to access agricultural trading and machinery sharing at the same platform on a single online platform. In most rural regions, farmers are not able to fetch good prices of their crops because of the role played by middlemen.

Simultaneously, not all farmers can afford to buy costly equipment, and renting machines can be hard to organize. KisanHaat hopes to overcome these two issues with a simple and practical web based solution. The site enables farmers to easily post and sell their farm produce to consumers without the need to use middlemen. Through this, farmers will be able to offer their own prices, communicate directly with buyers and have better profit margins. Instead, buyers are able to window shop items, find the best prices, and even shop fresh produce at the source.

In addition to trading agricultural goods, KisanHaat also provides a machinery rental feature. Farmers who own tractors, harvesters, or other farm equipment can list their machinery for rent. Other farmers in nearby areas can search for available equipment within a specific distance and book it for a required period. This not only reduces the financial burden of purchasing costly machinery but also ensures better utilization of existing resources.

The system works through a user-friendly web interface where users register and choose their role as either farmer or buyer. Farmers can upload product details or machinery listings, while buyers and fellow farmers can search based on location, price, and availability. The platform manages bookings, maintains transaction records, and ensures secure communication between users. An administrative panel monitors activities to maintain transparency and resolve disputes if required.

Overall, KisanHaat is more than just an online marketplace. It is a step toward digital empowerment in agriculture. By promoting direct trade and collaborative equipment sharing, the platform helps farmers increase income, reduce costs, and build a supportive agricultural community.

3.2 System Architecture

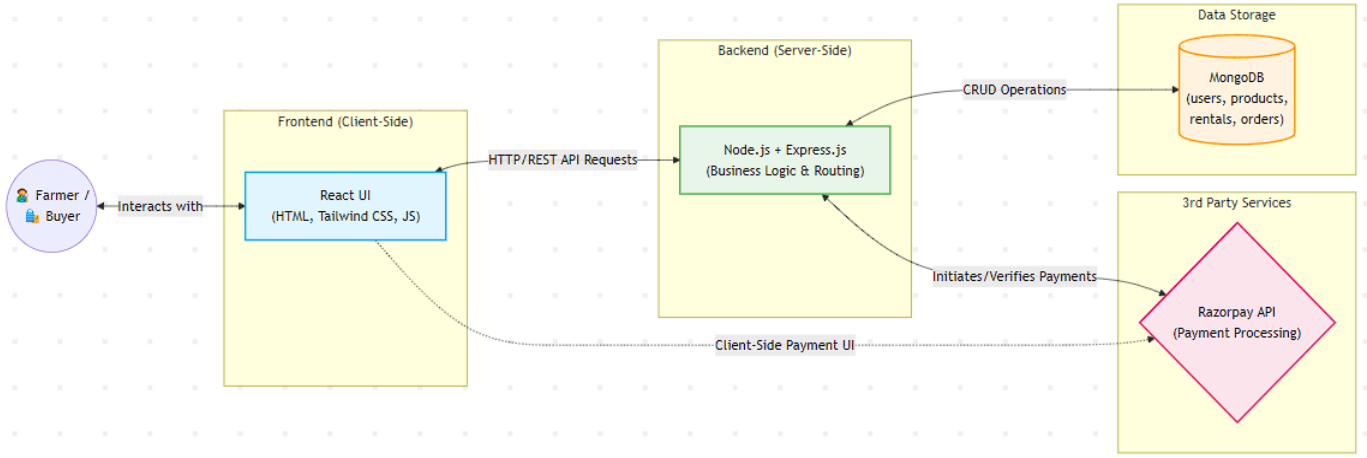

Fig: 1.1

Kisanhaat is based on modular 3-tier architecture that includes:

1. Presentation Layer (Frontend)
2. Application Layer (Backend)
3. Data Layer (Database)

3.2.1 Presentation Layer

React.js is used to develop the frontend of Kisanhaat. It provides:

- Registration and login page of users
- Product listing pages
- Machinery rental pages

- Cart management system
- Location-based filtering
- User dashboard

The interface is user-friendly and friendly to rural users.

3.2.2 Application Layer

The back-end is made with Express and Node.js. It handles:

- API requests
- Business logic
- Authentication and authorization
- Rental approval workflow
- Data validation
- Error handling

The backend provides good communication security between frontend and database.

3.2.3 Data Layer

The database is MongoDB that stores:

- User details
- OTP verification data
- Agricultural product listings
- Machinery listings
- Rental requests
- Transaction records

The use of MongoDB allows flexible schema design and efficient handling of dynamic marketplace data.

3.3 Working Methodology of Kisanhaat

The working of the proposed system can be explained in sequential steps:

Step 1: User Registration and Authentication

Users register using their phone number. OTP verification ensures secure login and prevent unauthorization access.

Step 2: Product Listing

The farmers will be able to add agricultural products by filling in information like:

Product name

Quantity

Price

Location

Description

The product is stored in the database and becomes visible to buyers.

Step 3: Machinery Listing

Machinery owners can list agricultural equipment for rent by specifying:

Machine name

Rental price per day

Availability

Location

Step 4: Product Search and Filtering

Products can be searched by buyers on the basis of:

Category

Price range

Location

Google Maps API is integrated to enable location-based filtering.

Step 5: Cart-Based Rental Request

Users can add machinery to their cart and send rental requests. This structured workflow ensures organized rental health service.

Step 6: Owner Approval Mechanism

The machinery owner receives the rental request and can:

Accept

Reject

Only after approval is the rental considered confirmed.

Step 7: Database Update

All transactions and rental records are securely stored in MongoDB for tracking and management.

3.3.1 Logistics and Supply Chain Workflow (Conceptual Model)

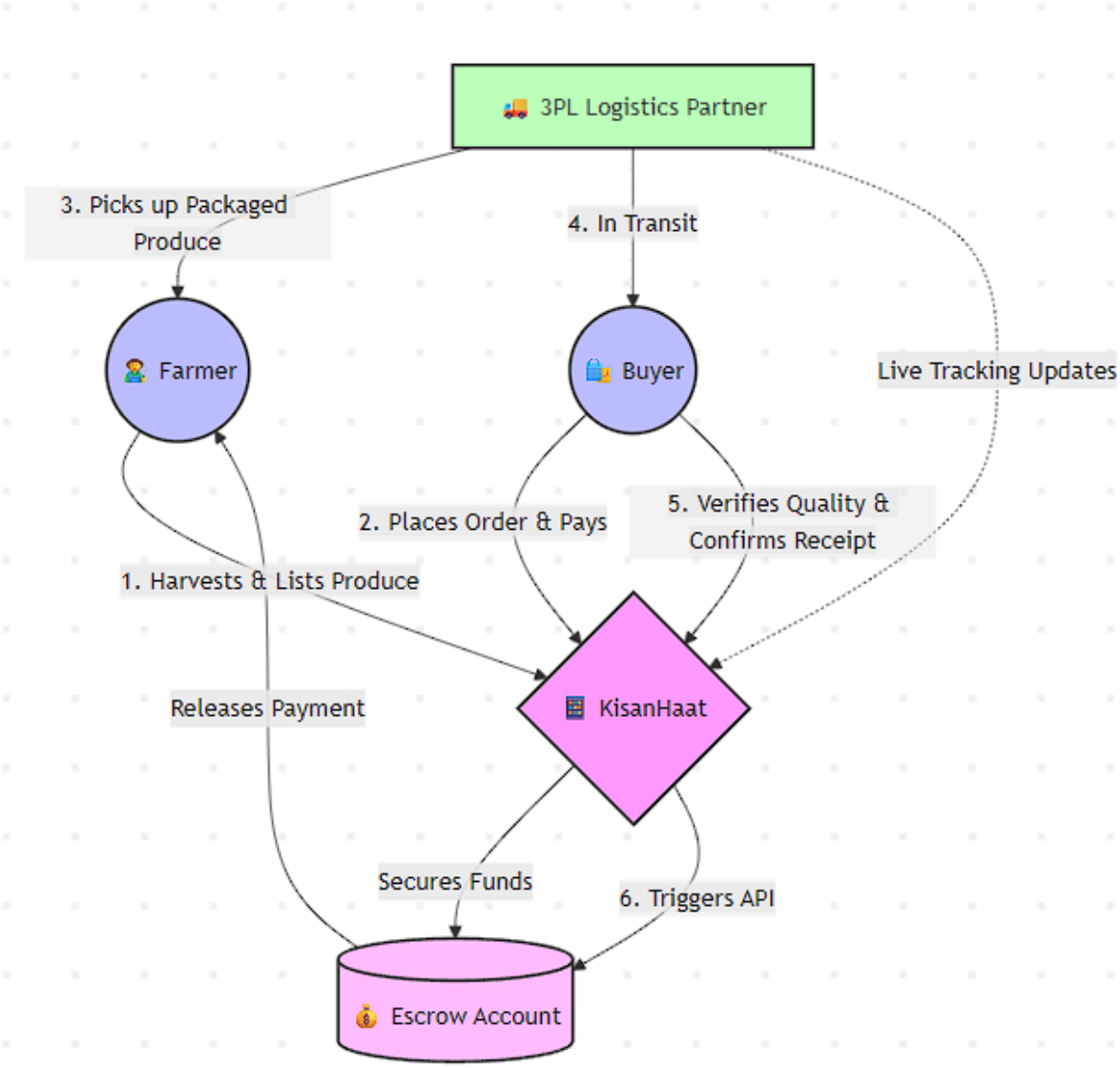

Fig: 1.2

The KisanHaat platform is made to facilitate a direct-to-buyer model of agricultural supply chain that minimizes the reliance on intermediaries and enhances transparency between farmers and buyers. The workflow of conceptual logistics is presented below.

1. **Harvest and Digital Listing (Farm Level)**

   The farmer carries out a rudimentary quality test after picking the farm produce. The product is further posted on the KisanHaat site with pictures posted, quantity supplied mentioned and a price anticipated.

2. **Place Order (Platform Level)**

   The consumers, restaurants, or retailers (the buyers) browse the offered products and can make orders via the platform. Order details are captured by the system and the farmer is informed of the purchase request.

3. **Packaging and Pickup (First-Mile Logistics)**

   After an order is confirmed, the farmer packages the produce according to some of the basic packaging guidelines. In the real life application, the produce may be picked at the farm or a collection point that is close to the farm by logistics partners.

4. **Transportation and Tracking (Mid-Mile Logistics)**

   Logistics services are used to deliver the produce to the purchaser. The platform can be used to track shipment status during transportation to facilitate transparency to the buyers and farmers.

5. **Delivery and Verification (Last-Mile Logistics)**

   The logistics partner will deliver the produce to the address of the buyer. Quality and quantity of the product are checked by the buyer on delivery.

6. **Settlement of Payments (Financial Flow)**

   Once the delivery has been confirmed; the system completes the payment and wires the sum to the farmer. This guarantees just pricing and safe monetary dealings between the purchaser and the farmer. This process flow illustrates how an online agricultural trading platform such as KisanHaat can streamline product trade, make supply chains more efficient, and decrease the involvement of the middlemen in agricultural markets.

3.4 Functional Modules of System

The Kisanhaat system consists of several interconnected modules:

3.4.1 Authentication Module

Handles phone number login and OTP verification to ensure secure access.

3.4.2 Product Management Module

It allows farmers to add, update, delete, and manage agricultural product listings.

3.4.3 Machinery Rental Module

Enables machinery owners to list equipment and manage rental requests.

3.4.4 Cart and Request Module

Manages cart functionality and structured rental request workflow.

3.4.5 Location-Based Search Module

Uses Google Maps integration to filter products and machinery by geographical location.

3.4.6 Database Management Module

Ensures efficient storage and retrieval of user, product, and rental data.

3.4.7 Admin/Monitoring Module

Tracks platform activity, manages users, and ensures smooth system performance.

3.5 Distinct Characteristics of the System

The following features differentiate Kisanhaat from traditional systems:

1. Integrated platform for both product trading and machinery rental.
2. Phone number-based OTP authentication for rural accessibility.
3. Cart-based structured rental request system.
4. Owner approval mechanism ensuring transparency.
5. Location-based filtering using mapping services.

6. Scalable full-stack architecture (React + Node + MongoDB).

7. User-friendly interface designed for agricultural users.

8. Secure database management with structured APIs.

9. Modular system design allowing future feature expansion.

10. Reduced dependency on intermediaries.

3.6 Benefits of Proposed System

The Kisanhaat system offers several advantages over traditional agricultural trading methods:

1. Increased Transparency – Direct interaction between farmers and buyers eliminates middlemen exploitation.

2. Improved Accessibility – Farmers can access wider markets digitally.

3. Efficient Machinery Utilization – Machinery owners can monetize idle equipment.

4. Secure Authentication – OTP-based login ensures safe user access.

5. Location-Based Convenience – Nearby listings reduce transportation costs.

6. Structured Rental Workflow – Owner approval ensures trust and accountability.

7. Scalability – System can support large numbers of users.

8. Cost-Effective – Reduces operational and transaction costs compared to physical mandis.

3.7 Feasibility Study (Technical, Economical, Operational)

There is a feasibility study that is conducted to know whether the proposed system is technically, economically and operationally feasible and practical. In Kisanhaat, three elements of feasibility have been well considered in order to assure successful implementation and practical application.

3.7.1 Technical Feasibility

The Kisanhaat system is technically viable since it is developed on well known and widely used web technologies like:

- React.js (Frontend)
- Node.js and Express.js (Backend)
- MongoDB (Database)

Such technologies are reliable, scalable and popular among the developers community.

No specialized hardware is needed in the system. It is able to operate effectively on normal computers, laptops and smartphones that have internet connectivity. The platform is web-based, and therefore, the user just needs a browser to log on to the system.

OTP authentication using phone number guarantees safe access without the use of complicated login details. Use of Google Maps API allows filtering of products and machinery based on location which enhances usability.

The server and the client use HTTPS and authentication middleware to secure data communication. Thus, the system is practically feasible in practice and can be used in real-life agricultural implementation.

3.7.2 Economical Feasibility

Kisanhaat is economically feasible because it is developed using open-source technologies. There is no dependency on expensive proprietary software or licensed platforms.

The project uses:

- Free development frameworks (React, Node.js)
- Open-source database (MongoDB Community Edition)
- Cloud deployment options with scalable pricing

Since agricultural users mainly access the platform through their smartphones, no additional hardware investment is required.

The modular architecture helps to minimize maintenance since there are separate components (frontend, backend, database) that can be fixed without influencing the whole system.

Moreover, Kisanhaat reduces operational costs for farmers by:

- Eliminating middlemen
- Reducing travel to physical markets
- Enabling direct buyer interaction
- Allowing machinery rental instead of purchasing costly equipment

Thus, the system is cost-effective for both developers and end users.

3.7.3 Operational Feasibility

Kisanhaat is developed to be easy to use. The interface is simple and user friendly, and it can be used even by technologically incompetent users.

Farmers can:

- Register using phone number
- List products easily
- Add machinery for rent
- Accept or reject rental requests

Buyers can:

- Search products
- Filter by location
- Add machinery to cart

- Send rental requests

The OTP-based login removes the need to remember passwords, increasing usability.

The system does not require extensive training. Basic smartphone and internet knowledge is sufficient to operate the platform. Therefore, Kisanhaat is operationally feasible and can be easily adopted in rural and semi-urban regions.

3.8 System Requirements

3.8.1 Functional Requirements

Functional requirements specify the main activities of the Kisanhaat system.

The system shall:

1. Allow users to register and login using phone-based OTP authentication.
2. Allow farmers to add, update, and delete agricultural product listings.
3. Allow machinery owners to list agricultural equipment for rent.
4. Enable buyers to search and filter products by category and location.
5. Provide cart functionality for machinery rental requests.
6. Allow machinery owners to accept or reject rental requests.
7. Store transaction and rental records securely in the database.
8. Provide location-based filtering using Google Maps integration.
9. Maintain user dashboard for managing listings and requests.

3.8.2 Data Requirement

The data requirements of the Kisanhaat system include structured marketplace data.

The system collects and stores:

1. **User Data**

    - Name
    - Phone number
    - Address
    - Role (Farmer / Buyer / Machinery Owner)

2. **Product Data**

    - Product name
    - Category
    - Quantity
    - Price
    - Description
    - Location

3. **Machinery Data**

    - Machine name
    - Rental price per day
    - Availability status
    - Location

4. **Rental Request Data**

- User ID
- Machine ID
- Rental duration
- Request status (Pending / Approved / Rejected)

All data is stored securely in MongoDB collections. Sensitive authentication data is protected and not exposed publicly.

3.8.3 Performance Requirements

The system must respond quickly to user actions without noticeable delay.

- Product search should return results within a few seconds.
- OTP verification must be processed instantly.
- The system must handle multiple users simultaneously.
- Database queries must be optimized for fast retrieval.

The platform should be scalable to support increasing numbers of users over time.

3.8.4 Maintainability Requirements

- Maintainability guarantees a long-term sustainability of the system
- Kisanhaat is based on a modular architecture
- Separate frontend and backend
- Structured API routes
- Organized database schema

This structure allows easy updates, bug fixing, and feature enhancement without disrupting the entire system.

3.8.5 Security Requirements

Security is critical because the system handles user information and transaction details.

The system implements:

- OTP-based authentication
- Encrypted communication using HTTPS
- Secure API endpoints
- Role-based access control

- Input validation to avert malicious attacks.

Data confidentiality and integrity of users are ensured throughout the interaction with the system.

3.9 Data Flow Diagram

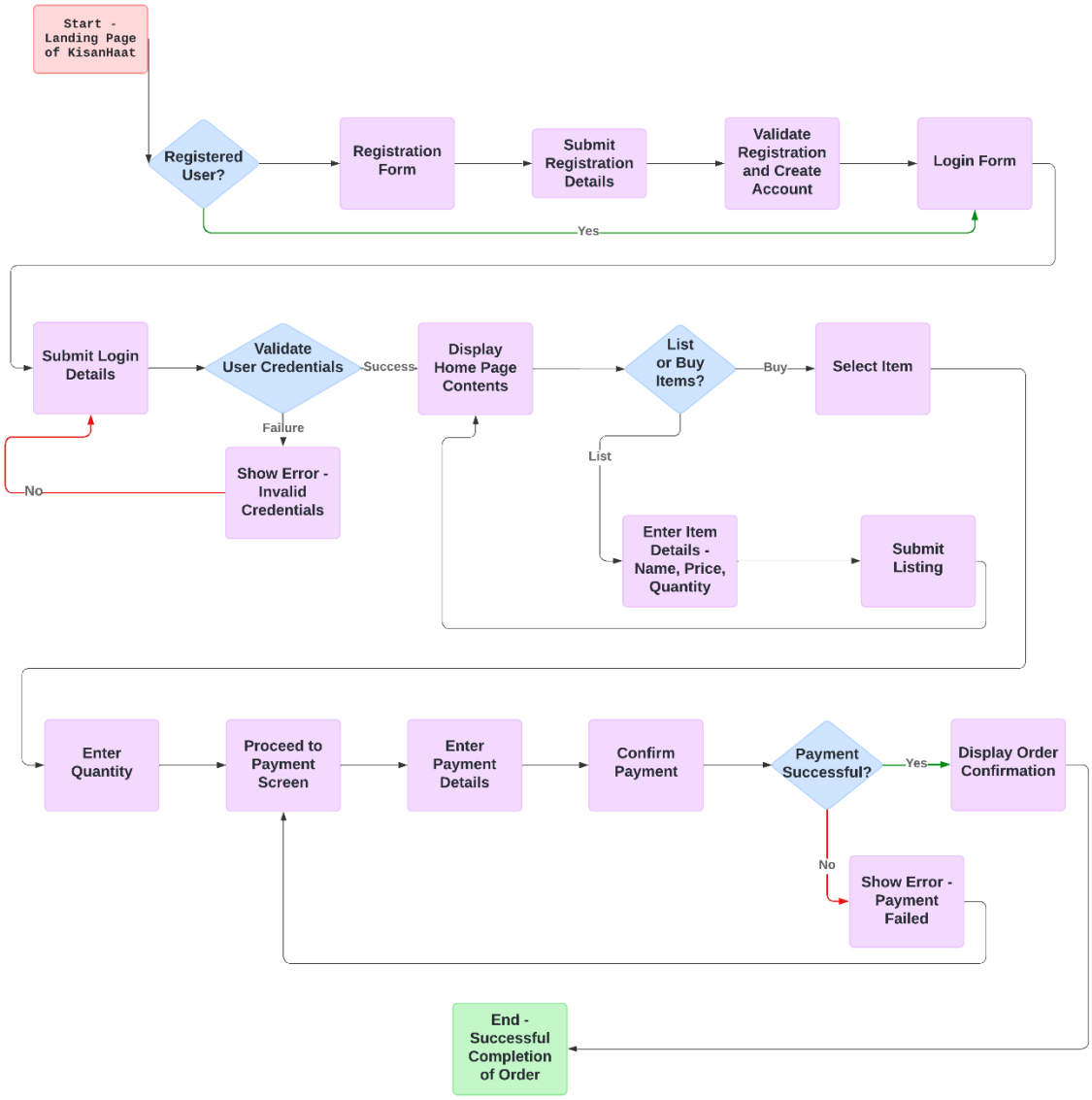

Fig:1.3

KisanHaat: An online platform of trading agricultural products and hiring machines.

The KisanHaat Data Flow Diagram shows how information flows among the users, system processes, and data storage elements. It describes the process through which the input data is received, processed, stored and returned as meaningful output within the platform.

1. External Entities

In the KisanHaat system, the primary external entity is:

User (Farmer / Buyer / Renter)

The user interacts with the system through a web interface. The user can:

Register a new account

Log into the system

List agricultural products or machinery

Browse and purchase items

Make online payments

Another external entity involved is:

Payment Gateway

This is an external service which handles financial transactions safely and sends transaction status (success or failure) to the system.

2. Main Processes in the System

The KisanHaat system consists of several major processes that transform input data into useful output.

Process 1: User Registration

This procedure gathers user information, including name, contact details and login.

The system checks the information entered so as to confirm accuracy and uniqueness.

In case of validation, user information is saved in the database.

In case of failure of validation, the system will inform the user to enter the details again.

Data Flow:

User Details → Validation Process → User Database

Process 2: User Authentication

During login, the system receives credentials from the user and verifies them against stored records.

If credentials match, access to the dashboard is granted.

If credentials do not match, an error message is generated.

Data Flow:

Login Credentials → Verification Process → User Database → Authentication Result

Process 3: Product Listing Management

This process enables users to upload details of agricultural products or machinery for sale or rent.

The system:

Accepts item name, price, quantity, and rental details

Validates input

Stores listing in the product database

Once stored, the listing becomes available for browsing by other users.

Data Flow:

Item Details → Validation → Product Database

Process 4: Product Browsing and Selection

Users have an ability to see items that are available. The system reads product details in the database and presents it in the interface.

Data Flow:

Product Database → Display Module → User

When a product is selected, the chosen item and quantity are forwarded to the order processing stage.

Process 5: Order Processing

This process handles order creation. It records:

Selected product

Quantity

Buyer details

The system temporarily stores order information before payment confirmation.

Data Flow:

Selected Item + Quantity → Order Processing → Order Database

Process 6: Payment Processing

This phase involves transfer of payment information to the third-party payment gateway.

In case of a successful transaction, the order status will be updated.

In case of failure, the system informs the user and lets him or her re-try.

Data Flow:

Payment Details → Payment Gateway → Transaction Status → Order Database

Process 7: Order Confirmation

Once payment verification has been successful, an order confirmation message is created by the system.

The confirmation details are retrieved from the order database and displayed to the user.

Data Flow:

Confirmed Order Data → Confirmation Module → User

3. Data Stores in the System

The main data repositories in KisanHaat include:

User Database – Contains information of registered users along with their login credentials.

Product Database – Contains all listed agricultural products and machinery details.

Order Database – Stores transaction records and order status.

Transaction Records – Maintains payment confirmation logs.

Each process interacts with these data stores to retrieve or update information as required.

4. Overall Data Movement Summary

The DFD represents a structured flow of information in the KisanHaat platform:

User provides input through registration, login, listing, or purchasing.

The system processes and validates the input.

Verified data is stored in appropriate databases.

When required, stored data is retrieved and displayed.

Payment transactions are securely handled through an external gateway.

Final output is delivered in the form of listing confirmation or order confirmation.

This structured data movement ensures that the platform maintains data integrity, transaction security, and smooth interaction between buyers and sellers. The DFD clearly shows how KisanHaat operates as an integrated digital marketplace supporting agricultural trade and machinery rental in an organized and secure manner.

3.10 Tools and Technologies Used

The implementation of Kisanhaat is successful because of integration of new full-stack web technologies that provide scalability, security and usability. The chosen technologies are used extensively in industry because of their performance, open-source and good community support.

The system is built based on a three-tier architecture of the frontend, backend, and database layers.

This separation ensures modular development and easier maintenance.

Frontend Development

React.js is used to develop the frontend of Kisanhaat. React was selected due to:

- It enables creation of reusable UI components.
- It provides fast rendering through virtual DOM.
- It supports dynamic data updates without reloading the page.
- It ensures a responsive and user-friendly interface.

The frontend handles:

- User registration and OTP login
- Product listing forms
- Machinery rental pages
- Cart management
- Dashboard display
- Location-based filtering interface

The design is kept simple and intuitive so that rural users can easily operate the platform.

Backend Development

The backend is done through the Express.js and Node.js. Node.js.

The choice of node.js was based on:

- It is efficient and lightweight.
- It supports asynchronous operations.
- It can be used in real time applications.

Express.js is used to:

- Create RESTful APIs
- Handle routing
- Manage middleware

- Process HTTP requests and responses

The tasks that the backend does are:

- OTP generation and verification
- User authentication and authorization
- Product and machinery management
- Rental request handling
- Owner approval workflow
- Database communication

Database Management

Kisanhaat uses **MongoDB** as the database.

MongoDB was selected because:

- It is a NoSQL database suitable for dynamic marketplace data.
- It supports flexible schema design.
- It provides high performance and scalability.

MongoDB stores:

- User details
- Product listings
- Machinery listings
- Rental requests
- Transaction records

Mongoose ORM is used to define schemas and interact with the database efficiently.

Authentication System

The system uses Phone Number OTP Authentication.

The process includes:

1. User types in phone number.
2. OTP is created and dispatched.
3. OTP is verified.
4. Access is granted upon successful verification.

This method is chosen because it is simple and convenient for agricultural users who may forget passwords.

Location Integration

Google Maps API is integrated to:

- Display product locations
- Filter products based on proximity
- Help users find nearby sellers or machinery

This feature improves usability and reduces transportation costs.

Deployment

The system can be deployed on cloud platforms such as:

- Node hosting services
- MongoDB Atlas for cloud database

The modular structure allows easy scaling and maintenance.

3.11 Dataset Description

Unlike machine learning systems, Kisanhaat does not rely on predictive datasets. Instead, it uses structured marketplace data generated dynamically by users.

The system database consists of the following collections:

1. User Collection

Stores information about registered users:

- User ID
- Name
- Phone number
- Address
- Role (Farmer / Buyer / Machinery Owner)

This data enables role-based functionality within the system.

2. Product Collection

Stores agricultural product details:

- Product ID
- Product name
- Category
- Quantity
- Price
- Description
- Location
- Seller ID

This allows buyers to search and purchase products efficiently.

3. Machinery Collection

Stores machinery rental details:

- Machine ID
- Machine name
- Rental price per day
- Availability status
- Location
- Owner ID

Important: Machines are rented, not sold.

4. Rental Request Collection

Stores rental transaction data:

- Request ID
- User ID
- Machine ID
- Rental duration
- Request status (Pending / Approved / Rejected)
- Timestamp

This collection supports the structured rental workflow.

Data Processing

The system processes user input data through:

- Input validation
- Form data sanitization

- API validation checks

All data is stored securely in MongoDB after validation.

Data Security

The system ensures:

- Secure authentication
- Encrypted communication (HTTPS)
- Protected API endpoints
- Role-based access control

Sensitive user information is not publicly exposed.

Summary of Implementation

The implementation of Kisanhaat successfully combines:

- Modern frontend technology (React.js)
- Efficient backend system (Node.js + Express)
- Scalable database (MongoDB)
- Secure OTP authentication
- Location-based filtering
- Cart-based rental workflow with owner approval

The system is scalable, secure, and user-friendly, so it is appropriate to use in the real-world agricultural implementation.

CHAPTER 4

RESULTS AND DISCUSSION

The testing is a critical step in the development of Kisanhaat to verify that the system is functioning properly, safely and effectively. Because Kisanhaat is a full-stack web application that includes user authentication, database operations, product listing and machinery renting work flows, a systematic testing was conducted in the various development phases.

Testing aims at detecting errors, enhancing the performance of the system as well as ensuring all the requirements (functional and non-functional) are met.

4.1 Testing Methodology

Different testing techniques were applied during development to ensure reliability and stability of the system.

4.2 Unit Testing

Individual modules of the Kisanhaat system were unit tested.

Each module was tested separately, such as:

- OTP authentication module
- Product listing module
- Machinery rental module
- Cart functionality
- Owner approval system
- Database operations

For example:

- OTP generation and verification were tested to ensure correct validation.
- Product form validation was tested to avoid empty or incorrect data submission.
- Rental request creation was tested to ensure proper database insertion.

Unit testing helped in identifying logical and coding errors at an early stage, reducing complexity during integration.

4.3 Integration Testing

After unit testing, integration testing was performed to verify the interaction between different modules.

Integration testing was the case in Kisanhaat:

- Easy interaction between frontend (React) and backend (Node.js).
- Handling API responses properly.
- Accurate data storage in MongoDB.
- Correct flow of rental requests from buyer to machinery owner.

Special attention was given to:

- OTP authentication flow
- Cart to rental request conversion
- Owner approval and status updates

This guaranteed the proper flow of data throughout the system, without loss or inconsistency.

4.4 Functional Testing

The functional testing of the system was done to ensure that it has fulfilled all the stipulated functional requirements.

Test cases included:

- User registration and login via OTP
- Adding agricultural products
- Editing and deleting listings
- Searching products by category and location
- Adding machinery to cart
- Sending rental requests
- Accepting or rejecting rental requests

Every functionality was evaluated in the eyes of the user to make sure that the system performs as anticipated.

4.5 Performance Testing

Performance testing was to be done to test the responsiveness and scalability of the system.

These aspects were put to test:

- Page loading time
- API response time
- OTP verification speed
- Database query performance
- Multiple user interactions

The system was loaded with moderate loads to ascertain that it is capable of supporting multiple users without any significant lag.

4.6 Security Testing

Security testing is critical because Kisanhaat handles user data and transaction information.

Security measures tested include:

- OTP-based authentication validation
- Role-based access control
- Secure API endpoints
- Prevention of unauthorized access
- Input validation to avoid injection attacks

Encrypted communication (HTTPS) was verified to ensure secure data transmission between client and server.

4.7 System Testing

After completing unit and integration testing, full system testing was performed.

This involved testing the entire workflow:

1. User registers using phone number.
2. OTP verification is completed.
3. Farmer lists agricultural products.
4. Buyer searches and views products.
5. Machinery is added to cart.
6. Rental request is sent.
7. Owner approves or rejects request.

8. Database updates reflect the changes.

This confirmed that the complete system works smoothly in real-world scenarios.

4.8 Maintenance of the System

Maintenance is an ongoing process that ensures the system continues to operate efficiently after deployment.

Kisanhaat is developed using a modular architecture, which makes it easy to update or modify specific components without affecting the entire system.

Maintenance activities include:

4.8.1 Corrective Maintenance

Corrective maintenance is carried out to correct bugs or errors that are detected during deployment.

Examples include:

- Fixing OTP validation issues
- Correcting database errors
- Resolving API routing problems
- Fixing UI responsiveness issues

This ensures uninterrupted service for users.

4.8.2 Perfective Maintenance

Perfective maintenance is aimed at enhancing the performance and user experience of the system.

Improvements may include:

- Enhancing search filters
- Improving UI design

- Optimizing database queries
- Reducing page load time
- Improving dashboard visualization

User feedback plays an important role in guiding such improvements.

4.8.3 Preventive Maintenance

Preventive maintenance is performed to prevent failures of the system in the future.

This includes:

- Regular security updates
- Database backup
- Monitoring server performance
- Updating dependencies and libraries
- Testing API endpoints periodically

These steps ensure long-term stability, security, and scalability of Kisanhaat.

4.9 Processed Data Handling in Kisanhaat

In Kisanhaat, user inputs such as product details, machinery information, and rental requests are processed before being stored in the database.

The processing includes:

- Input validation
- Data sanitization
- Format normalization
- Role verification

This ensures that:

- Invalid data is rejected
- Database integrity is maintained
- Security risks are minimized

The structured handling of processed data makes the system scalable and maintainable. Since the backend APIs follow standardized data formats, future upgrades (such as adding online payments or review systems) can be implemented without restructuring the entire system.

Conclusion of Testing and Maintenance

Through comprehensive testing methodologies and continuous maintenance strategies, Kisanhaat ensures:

- Stability
- Security
- Performance efficiency
- User satisfaction
- Long-term sustainability

The structured testing process validates both technical and functional reliability, while the maintenance strategy guarantees that the platform remains adaptable to future technological advancements and increasing user demands.

As a result, Kisanhaat is a dependable and scalable agricultural marketplace system suitable for real-world deployment.

4.10 Result Analysis

This chapter shows the findings of the implementation and testing of the Kisanhaat system. Result analysis is aimed at testing the functionality, efficiency, performance, security and scalability of the system under real conditions of use.

The evaluation of Kisanhaat was carried out by simulating different user roles:

- Farmers listing agricultural products
- Buyers searching and purchasing products
- Machinery owners listing equipment for rent
- Users sending rental requests
- Owners approving or rejecting requests

The system was subjected to different usage conditions to check on stability and reliability.

The findings are displayed in tabular format, graphical and system workflow observations.

Functional Result Analysis

The primary objective of Kisanhaat is to provide:

1. A marketplace for buying and selling agricultural products.
2. A platform for renting agricultural machinery.

The following outcomes were observed:

Product Listing Module

- Farmers were able to successfully add product details.
- Product data was stored correctly in MongoDB.
- Buyers were able to search and filter products.
- Location-based filtering using Google Maps worked accurately.

Machinery Rental Module

- Machinery owners successfully listed equipment.

- Users added machines to cart without errors.
- Rental requests were created properly in the database.
- Owner approval and rejection functionality worked correctly.
- Rental status updated dynamically on the user dashboard.

This will ensure that the system satisfies all the functional requirements.

4.11 Performance Evaluation

Performance evaluation was conducted to assess:

- Response time
- API efficiency
- Database query performance
- Scalability under multiple users

Response Time

The average response time for:

- Login with OTP: Immediate verification within seconds
- Product search: Fast retrieval from database
- Rental request submission: Successfully processed without delay

The system maintained consistent performance during multiple simultaneous user interactions.

Database Performance

MongoDB handled:

- Multiple product listings
- Concurrent rental requests

- Frequent updates to rental status

Query optimization ensured quick retrieval of product and machinery data.

Scalability

The system was tested with increasing numbers of:

- Product listings
- Machinery listings
- User accounts

Results showed that the modular architecture supports scalability without significant performance degradation.

Since Node.js supports asynchronous processing, the system handled concurrent requests efficiently.

Security Evaluation

Security testing validated the following:

- OTP-based authentication prevented unauthorized login.
- Role-based access control restricted system functions appropriately.
- API endpoints were protected from unauthorized access.
- User data remained confidential and secure.

The system ensured secure communication through HTTPS protocols.

4.12 Discussion

The implementation and testing of Kisanhaat provided several important insights.

1. Direct Farmer-to-Buyer Interaction

The system is able to remove reliance on go-betweens because it allows direct contact between the sellers and buyers.

This enhances the price transparency and enhances the profitability of the farmers.

2. Efficient Machinery Utilization

The machinery rental module allows owners to earn income from idle equipment.

Users can rent machinery instead of purchasing expensive equipment, reducing financial burden.

3. Structured Rental Workflow

The cart-based rental request system ensures organized transaction flow.

The owner approval mechanism increases trust and prevents misuse.

4. Location-Based Accessibility

Google Maps integration enhances usability by allowing users to find nearby products and machinery.

This reduces transportation cost and time.

5. System Reliability

The system remained stable under different user loads.

The modular full-stack architecture ensures long-term maintainability and flexibility.

6. Practical Applicability

Unlike theoretical systems, Kisanhaat is practical and suitable for real-world agricultural communities.

The web-based nature of the platform ensures accessibility using smartphones and basic internet connectivity.

Overall Discussion

The experimental evaluation demonstrates that Kisanhaat is:

- Functionally complete
- Performance-efficient
- Secure
- Scalable
- User-friendly

The platform successfully integrates agricultural product trading and machinery rental within a single digital marketplace.

The modular design allows future enhancements such as:

- Online payment integration
- Review and rating systems
- Multilingual support
- Government scheme integration
- Real-time chat between users

The findings support that Kisanhaat satisfies the functional and non-functional requirements and could be deployed in the real world.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

KisanHaat: A Web-Based Platform for Agricultural Product Trading and Machinery Rental represents an effort to apply modern web technologies to address real challenges in the agricultural sector. The system was designed to create a digital platform where farmers, buyers, and machinery owners can interact directly in a secure and organized environment. By reducing dependency on intermediaries and improving transparency, the platform simplifies agricultural product trading and machinery rental processes.

During the development of the system, emphasis was placed on designing a structured and reliable architecture. The platform integrates frontend interfaces, backend services, and database management in a well-organized manner. The main modules like user authentication, product listing, machinery rent management, and order handling were introduced in order to facilitate a smooth interaction between the users and the system.

The system also ensures secure data handling and consistent transaction flow. All operations such as adding products, renting machinery, and processing user requests follow clearly defined workflows supported by proper validation and error handling mechanisms. CRUD operations were successfully implemented using MongoDB and Express.js, demonstrating efficient database interaction and dynamic data management.

In general, KisanHaat is a viable and scalable solution to agricultural trading and machinery sharing digitalization. The project shows how an organized web-based system can enhance access, transparency, and efficiency in agricultural markets.

5.2 Future Scope

Although the current system fulfills its primary objectives, several enhancements can further improve its functionality and real-world usability.

One of the major future improvements is the implementation of a **location-based machinery rental system within a 50 km radius** of a specific city. This feature will allow users to view only those machines that are geographically accessible, reducing transportation difficulties and making the rental process more practical. Distance calculation logic and optimized database queries can be integrated to support efficient proximity-based filtering.

A second consideration which is worth implementing is a real-time notification system. Users are able to get instant notifications when new machinery is available in their locality, when a request to rent is approved or denied, or when something significant happens. This aspect will enhance the interaction of the platform.

Future versions of the system may also include a **review and rating mechanism** that allows users to evaluate machinery owners and buyers. This will help build trust and transparency within the platform. Additionally, an **analytics dashboard** can be introduced to monitor user activity, transactions, and marketplace trends.

Further improvements may include stronger **role-based access control**, integration of **intelligent recommendation systems**, and deployment on **scalable cloud infrastructure** to support a larger number of users. Advanced security measures such as **multi-factor authentication** can also be implemented to enhance system protection.

With these improvements, KisanHaat has the potential to evolve into a more comprehensive digital ecosystem that supports efficient agricultural trading and machinery sharing for the farming community.

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